Systems Analysis and Control

Matthew M. Peet
Arizona State University

Lecture 17: Lead-Lag and Notch Filter Design with Root Locus

Overview

In this Lecture, you will learn:

Lead-Lag Compensation

- Designing Leads
  - Pole Placement using root locus
- Designing Lags
- Combining Leads and Lags

Notch Filters

- Providing complex zeros at targeted Locations
- Changes Departure Angles.

Recall: Pole-Zero Compensation

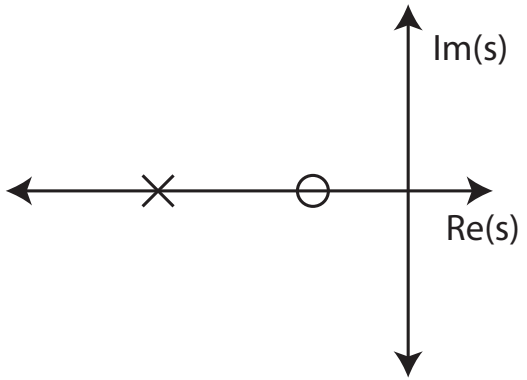
Definition 1.

A **Pole-Zero Compensator** is of the form

$$K(s) = \frac{s+z}{s+p}$$

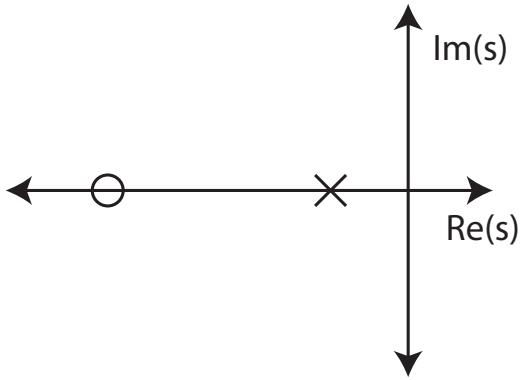
Lead Compensation

- $p < z$
- Replaces Pure Zero
- Improves Root Locus



Lag Compensation

- $z < p$
- Replaces Integrator
- Reduces Steady-State Error



Lead Compensation for Inverted Pendulum

Effect of Lead Compensator on the Root Locus

$$G(s) = \frac{1}{s(s+1)}$$

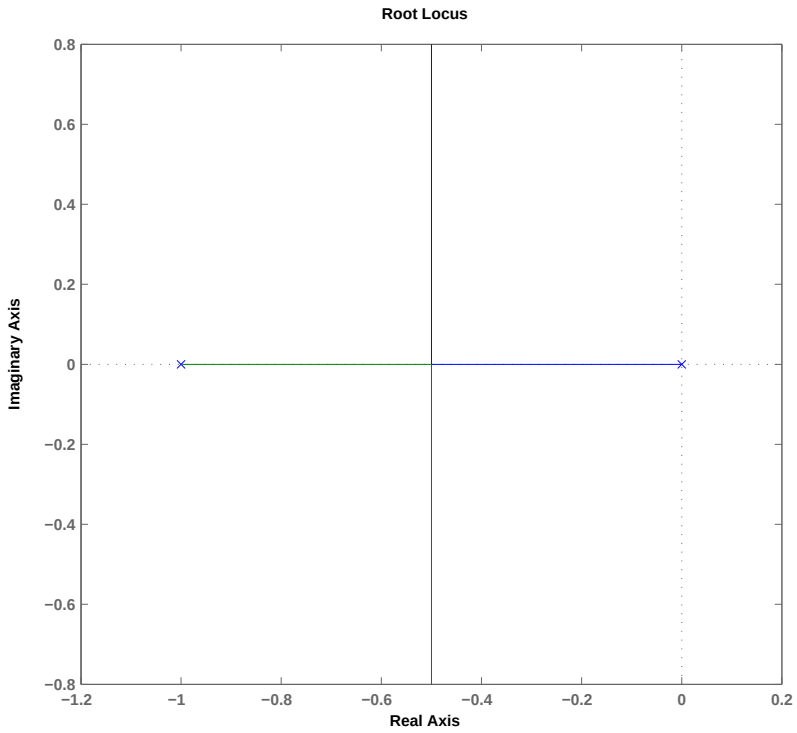
Asymptotes: $\pm 90^\circ$

Intercept: $\alpha = -.5$

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m} = \frac{-1-0}{2-0} = -.5$$

Break Point: $s = -.5$

$$n'd - d'n = 2s + 1 = 0$$



Conclusion: At high gain, we get

- High Frequency Oscillation
- Lots of overshoot
- Fixed Settling Time

Lead Compensation for Inverted Pendulum

Effect of Lead Compensator on the Root Locus

Asymptotes: $\pm 90^\circ$
Intercept: $\alpha = -1$

$$\alpha = \frac{\sum p_i - \sum z_i}{n - m}$$
$$= \frac{-5+4}{2-0} + \frac{-1-0}{2-0} = -.5 - .5 = -1$$

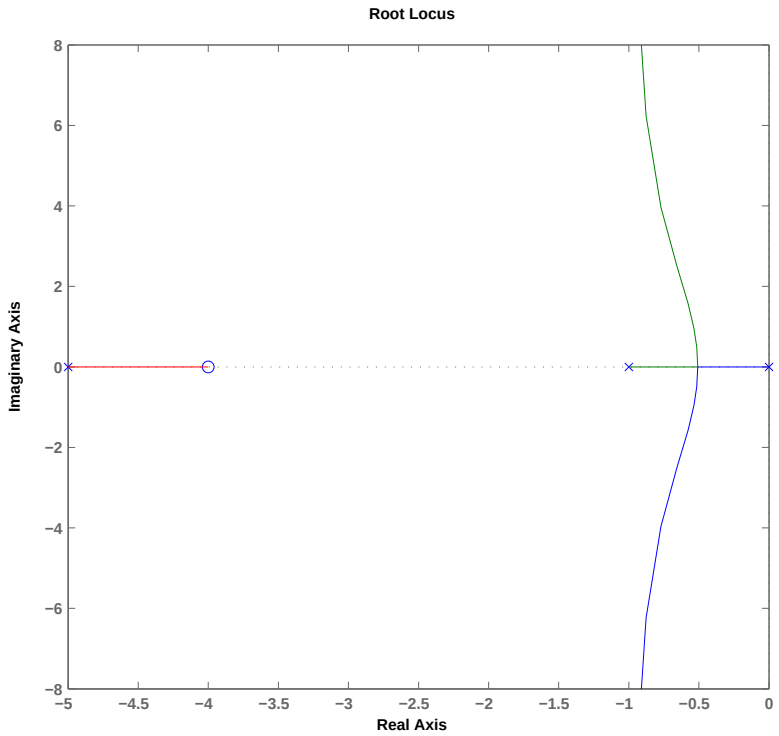
Break Point: $s = -.508$

$$nd' - n'd$$
$$= (s+5)(s^2+s) - (3s^2+12s+5)(s+4)$$

Conclusion: At high gain, we get

- Improved Settling time
- Slightly less overshoot

$$G_c(s) = \frac{s+4}{s+5}$$



Lead Compensation Increases Phase

Moves Root Locus to the Left

The effect of a **Lead Compensator**

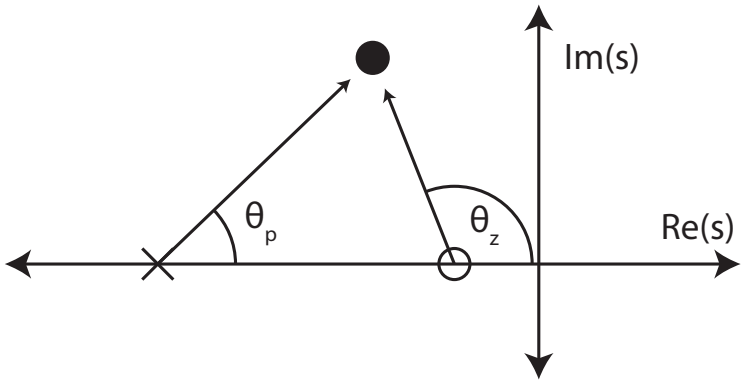
- Add Phase at every point

$$\angle(K(s)G(s)) = \angle K(s) + \angle G(s)$$

- The change in phase is positive.

$$\Delta\angle = \theta_z - \theta_p$$

Root Locus compensates by moving left.



Lead Compensation Can be Used for Pole Placement

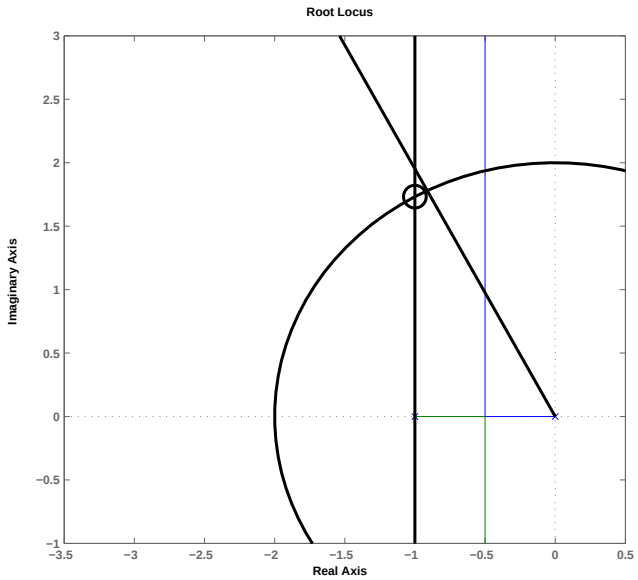
Make the Root Locus Pass through a Specific Point

Lead-Lag can be used to do pole-placement

$$G(s) = \frac{1}{s(s+1)}$$

Suppose we want:

- 20% Overshoot
- $\omega_n = 2$
- $T_s < 4$



Choose a desired pole location:

- The intersection of
  - $\omega_n = 2$
  - $\sigma < -1$

$$s_{1,2} = -1 \pm \sqrt{3}i$$

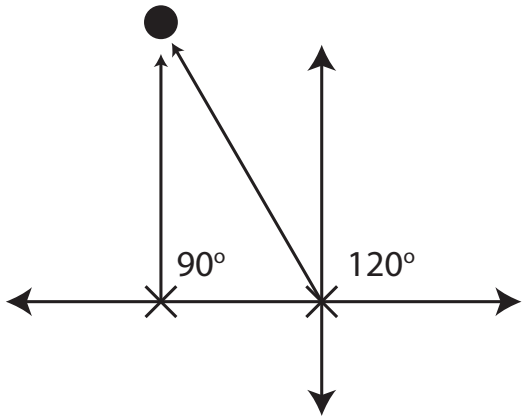
Question: Can we force the root locus to pass through this point using Pole-Zero compensation?

Pole Placement Using Lead Compensation

Choose a Desired Point and Compute the Phase

Lets start with a basic question:

- Is $s = -1 \pm \sqrt{3}i$ already on the root locus?



Lets check:

$$\angle G(s) = \sum \angle(s - z_i) - \sum \angle(s - p_i)$$

Working out the geometry:

$$\angle G(s) = -90^\circ - 120^\circ = -210^\circ$$

Not on the Root Locus!

The point s lacks 30° of phase.

Pole Placement Using Lead Compensation

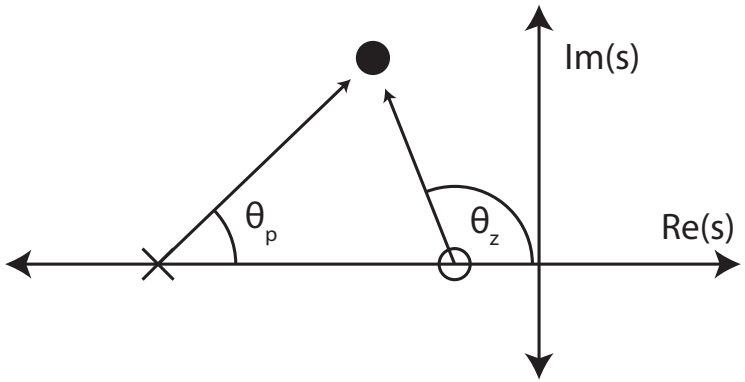
Make the Phase at Desired Point 180°

To place the point $s = -1 \pm \sqrt{3}i$ on the root locus:

- we need to add 30° of phase at this point.

Phase is sum of zeros minus poles

- Zeros add phase
- Poles subtract phase.



We can add 30° if we use a pole-zero combo:

- Add a zero at $\theta_z = 60^\circ$
- Add a pole at $\theta_p = 30^\circ$

Pole Placement Using Lead Compensation

Compute Pole-Zero Locations to Add the Required Phase

Desired point: $s = -1 \pm \sqrt{3}\imath$

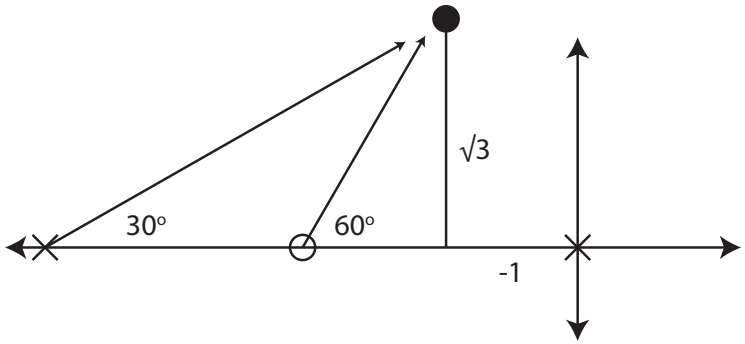
$$K(s) = \frac{s - z}{s - p}$$

Use reverse geometry to find p and z so $\theta_z = 60^\circ$ and $\theta_p = 30^\circ$.

Zero:

$$\tan 60^\circ = \frac{\sqrt{3}}{x}$$

$$x = \frac{\sqrt{3}}{\tan 60^\circ} = 1$$



Pole:

$$\tan 30^\circ = \frac{\sqrt{3}}{x}$$

$$x = \frac{\sqrt{3}}{\tan 30^\circ} = 3$$

$$p = -1 - x = -4$$

$$z = -1 - x = -2$$

Pole Placement Using Lead Compensation

Final Step: Compute Gain and Construct Controller

Now, the root locus passes through $s = -1 \pm \sqrt{3}i$.

To find the gain at this point

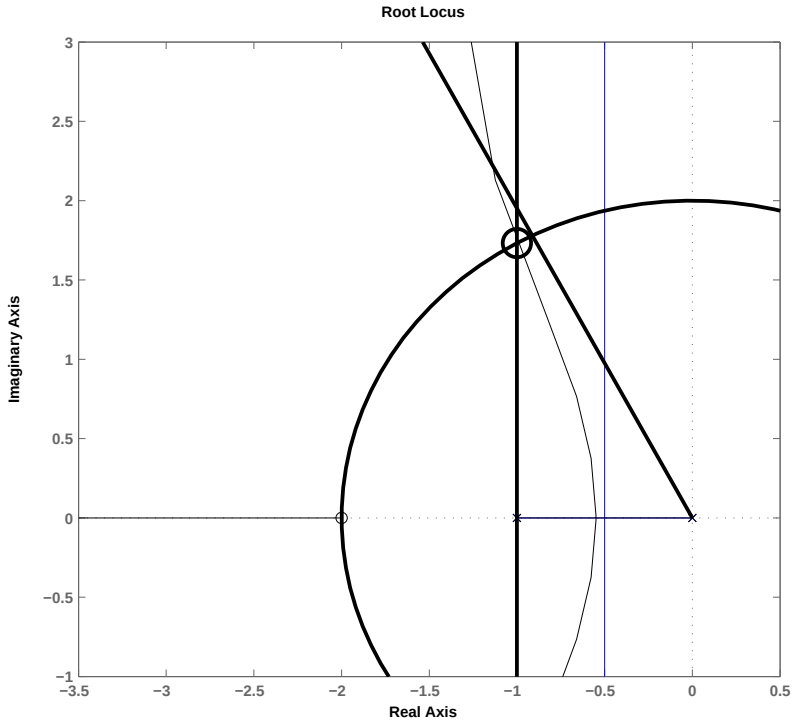
- Use `rlocfind`
- Use $k = \frac{|d(s)|}{|n(s)|}$.

For this example,

$$k = 6.00$$

Designed Controller:

$$K(s) = 6\frac{s/2+1}{s/4+1} = 12\frac{s+2}{s+4}$$



Potential Problem: May adversely affect other poles.

Changing Departure Angles Using Lead Compensation

Departure Angles Should Point Towards the Left!

The other big use of lead compensation is to change **Departure Angles**.

Recall the Suspension system problem with integral feedback:

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1} \frac{1}{s}$$

The poles are

- $p_{1,2} = -.957 \pm 1.23\imath$
- $p_{3,4} = -.0433 \pm .641\imath$

At pole $p_3 = -.0433 + .641\imath$, the phase is -156°.

Departure Angle:

$$\angle_{dep} = \angle G(p_3) + 180^\circ = 24^\circ$$

Goal: Increase the departure angle to 90° or more.

Changing Departure Angles Using Lead Compensation

Changing the Phase at an Open-Loop Pole Changes the Departure Angle!

The **Phase** at $p_3 = -.0433 + .641\imath$ is currently

$$\angle G(p_3) = -156^\circ$$

The **Departure Angle** from $p_3 = -.0433 + .641\imath$ is currently

$$\angle_{dep} = \angle G(p_3) + 180^\circ = 24^\circ$$

Lets make the departure angle $\angle_{dep} = 100^\circ$ using a Lead Compensator.

- How much phase do we need at p_3?

$$\angle_{req} G(p_3) = \angle_{dep} - 180^\circ = -80^\circ$$

- How much phase do we need **TO ADD** at p_3?

$$\Delta\angle G(p_3) = \angle_{req} G(p_3) - \angle G(p_3) = -80 + 156^\circ = 76^\circ$$

Changing Departure Angles Using Lead Compensation

Add the Required Phase using a Pole-Zero Compensator!

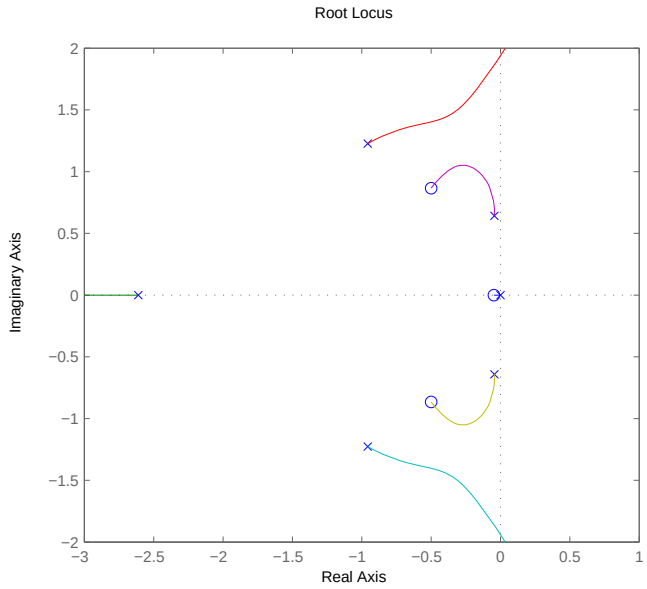
We need to add $76°$.

- Zero at $90°$.
- Pole at $14°$.

Recall departure point is $p_3 = -.0433 + .641\imath$

Zero:

- $\theta = 90°$,
- $\Delta x = 0$
- $z = -.0433$



Pole:

$$\tan 14° = \frac{.641}{\Delta x}$$

$$\Delta x = \frac{.641}{\tan 14°} = 2.57$$

So $p = -.0433 - \Delta x = -2.61$.

Controller:

$$K(s) = \frac{s + .0433}{s + 2.61}$$

Reducing Steady-State Error Using Lag Compensation

Reduces Steady-State Error by Factor p/z!

Predict the Steady-State Error.

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1} \qquad K(s) = \frac{s - z}{s - p} = \frac{n_K(s)}{d_K(s)}$$

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} \frac{1}{1 + G(s)K(s)} \\ &= \frac{d_K(0)}{d_K(0) + k n_K(0) G(0)} \\ &= \frac{-p}{-p + -zkG(0)} \end{aligned}$$

If

- p is small
- z is large

Then

$$e_{ss} \cong \frac{p}{kz} \frac{1}{G(0)}$$

Lag Compensation Messes up the Root Locus

To Reduce this Effect, Put the Lag Compensator far to the Left

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1}$$

Say we want steady-state error less than .01.

$$e_{ss} \cong \frac{p}{kz}\frac{1}{G(0)} = \frac{p}{kz} \leq .01$$

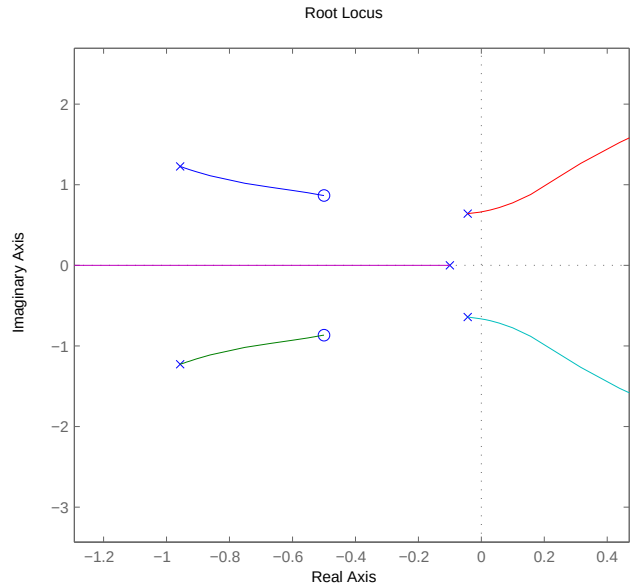
or $p \leq .01kz$

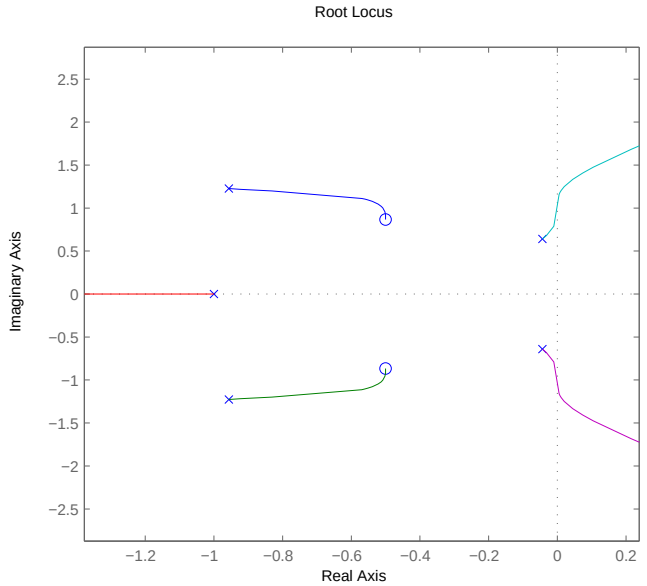
- Assume $k > 10$
- Choose $p = .1$
- Result: $z = 100$

$$K_{lag}(s) = \frac{s + 100}{s + .1}$$

Alternatively, $p = 1$, $z = 1000$.

- But there are dangers!





Lag Compensation Shifts Asymptotic Centers to the Right

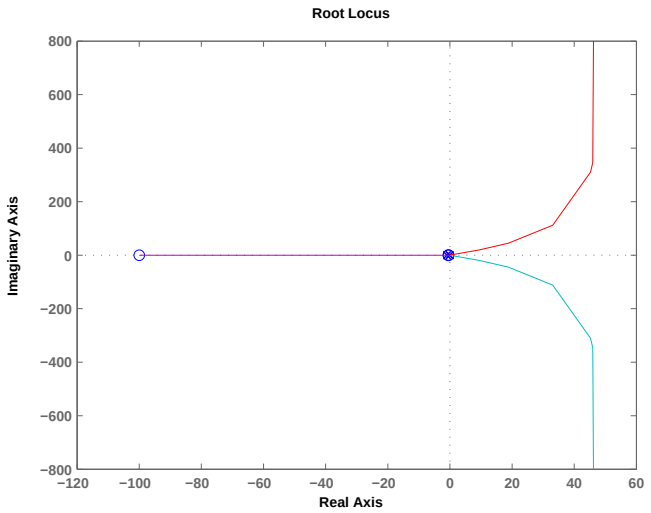
To Reduce this Effect, Keep the Gain Small

Notice some negative effects of Lag

- Asymptotes still at $\pm 90^\circ$

Center of Asymptotes:

$$\begin{aligned}\alpha &= \frac{\sum p_i - \sum z_i}{n-m} \\ &= \frac{\sum p_{i,old} - \sum z_{i,old}}{n-m} + \frac{\sum p_{i,new} - \sum z_{i,new}}{n-m} \\ &= \alpha_{old} + \frac{p-z}{2}\end{aligned}$$



Creates a Shift in Asymptotes by

$$\Delta\alpha \cong \frac{z}{2} = 50$$

for the suspension problem ($z = -100$).

Lead-Lag Compensation

Lead Compensation Mitigates the Negative Effects of Lag Compensation

To mitigate the effect of lag compensation:

- Add some Lead Compensation
  - Zero at $z = .01$
  - Pole at $p = 20$

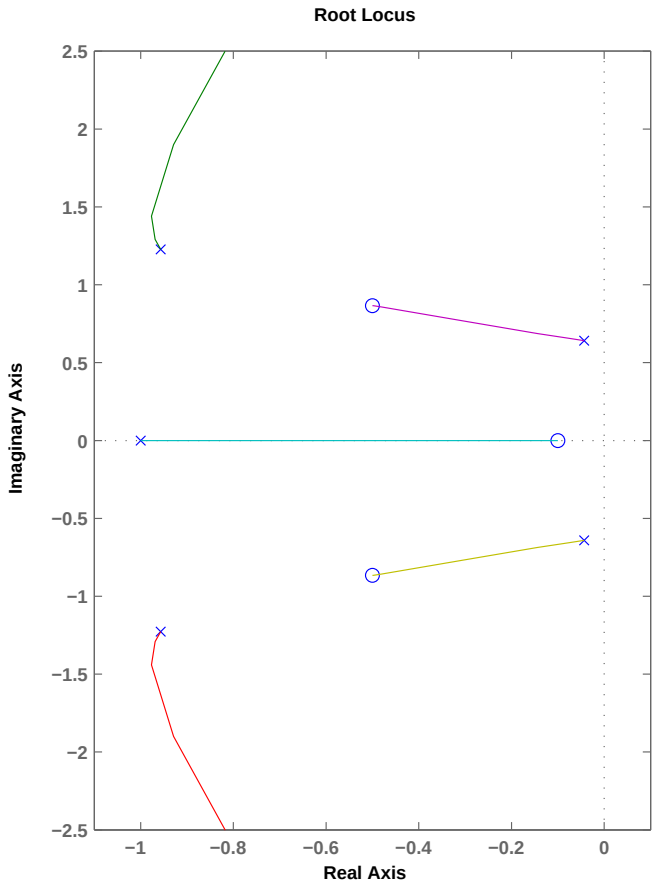
$$K_{lead}(s) = \frac{s + 100}{s + .1}$$

Phase at $p_3 = -.0433 + .641\imath$

$$\angle G(p_3) = -25.8^\circ$$

Departure Angle:

$$\angle_{dep} = 180 + \angle G(p_3) = 154.25^\circ$$

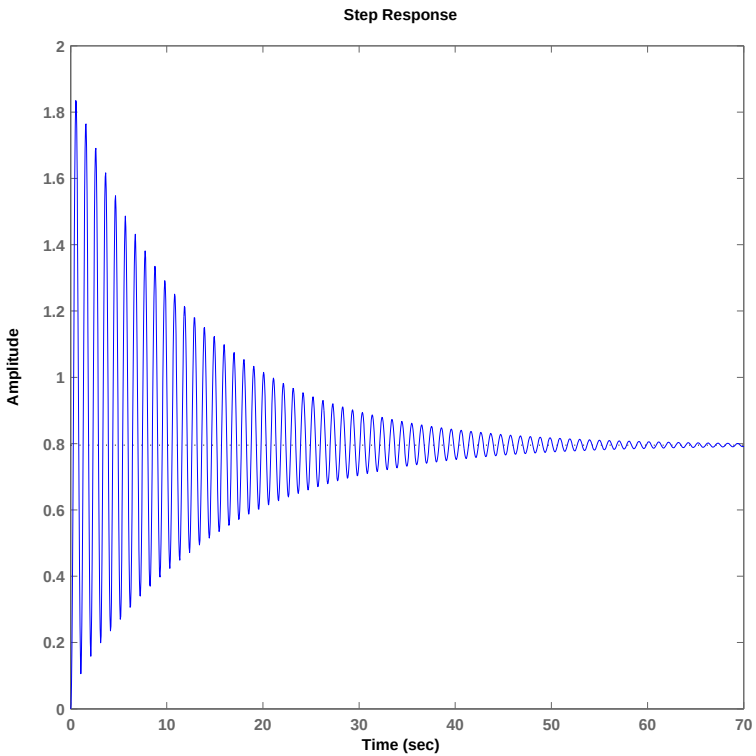


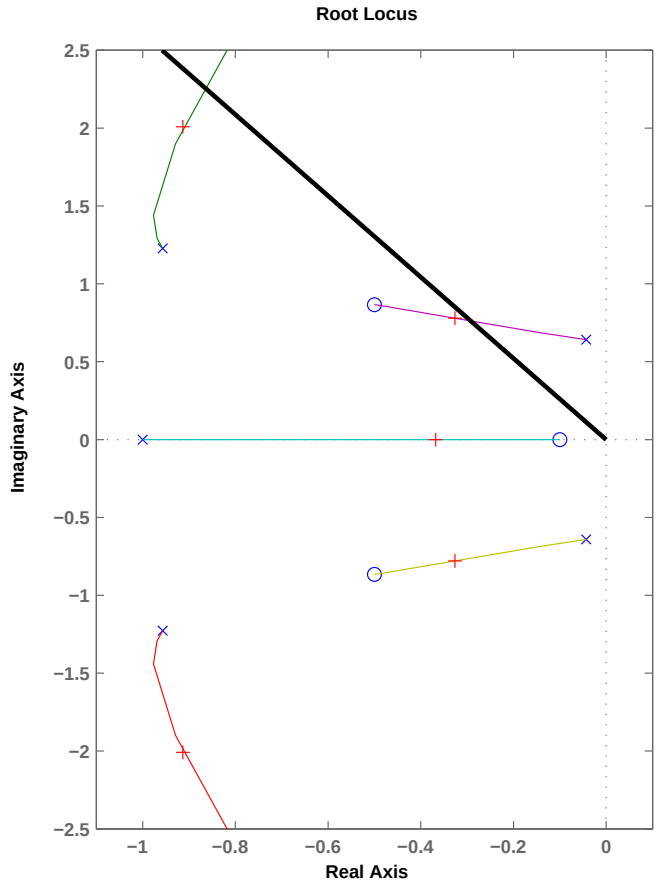
Combined Lead-Lag Compensation

Use `rlocfind` to pick off

- Maximum stable gain $k = .7768$
- Remember to normalize!

$$K_{lead-lag}(s) = .7768 \frac{.2}{1} \frac{s+100}{s+.1} \frac{s+.01}{s+20}$$





Notch-Filters Are Used to Change Departure Angles

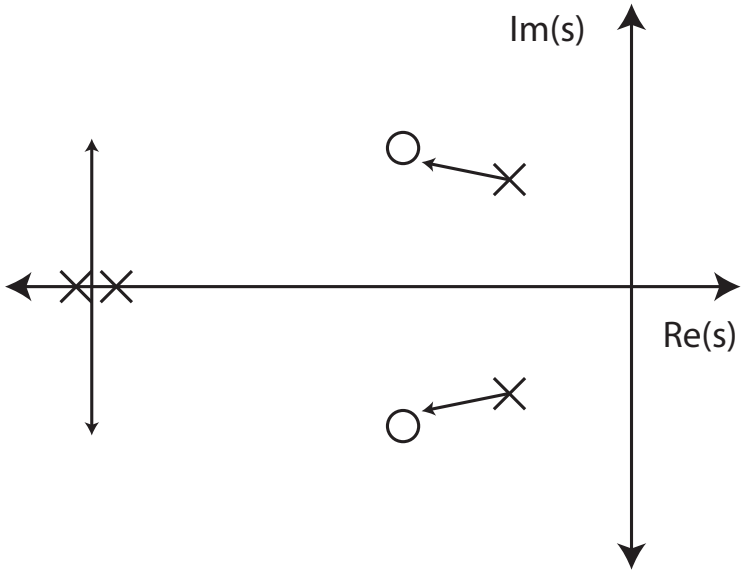
Use Complex Zeros to Target Phase change at Problematic Poles

Conclusion:

- Lead-Lag Improves Performance
- Limited to poles-zeros on real axis

Problem Can't Stabilize those poles at

$$s = -.0433 \pm .641\imath$$



One solution is to use a *Notch Filter*.

Definition 2.

A **Notch Filter** consists of

- Two Complex Zeros
  - Used to Capture Troublesome Poles
- Two Real Poles far out in the LHP

Using Notch-Filter to Target Open-Loop Poles

Change the Departure Angle

To attack the poles at

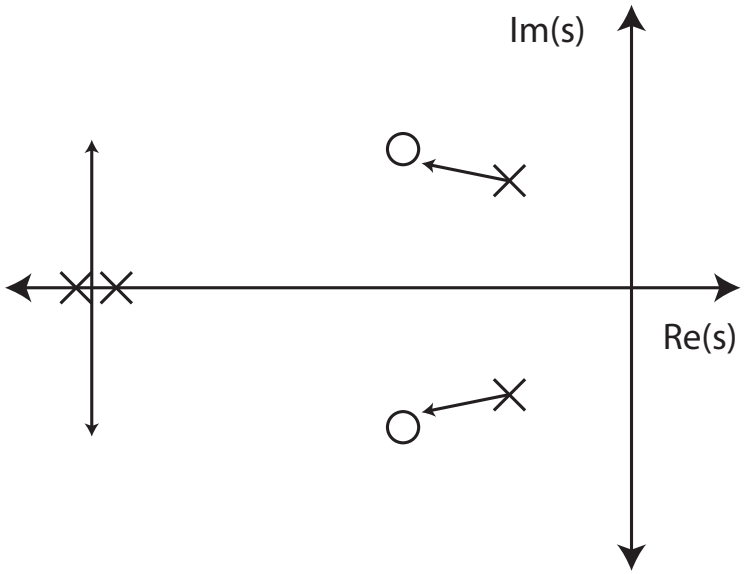
$$s = -.0433 \pm .641\imath$$

Lets use a notch filter at

$$z_{1,2} = -.5 \pm .641\imath$$

with poles way out at

$$p_{1,2} = -20$$



$$K(s) = \frac{(s + .5 + .641\imath)(s + .5 - .641\imath)}{(s + 20)^2}$$

$$= \frac{s^2 + s + .66}{s^2 + 40s + 400}$$

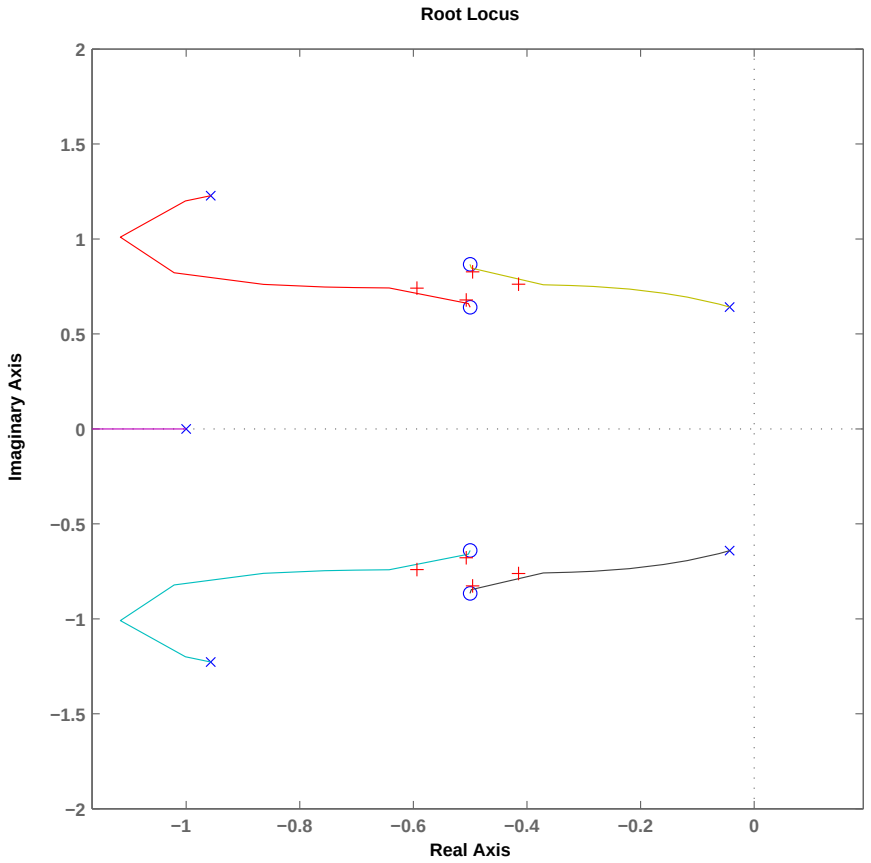
Notch-Filter Example

Combining the Notch-Filter with the Lag filter.

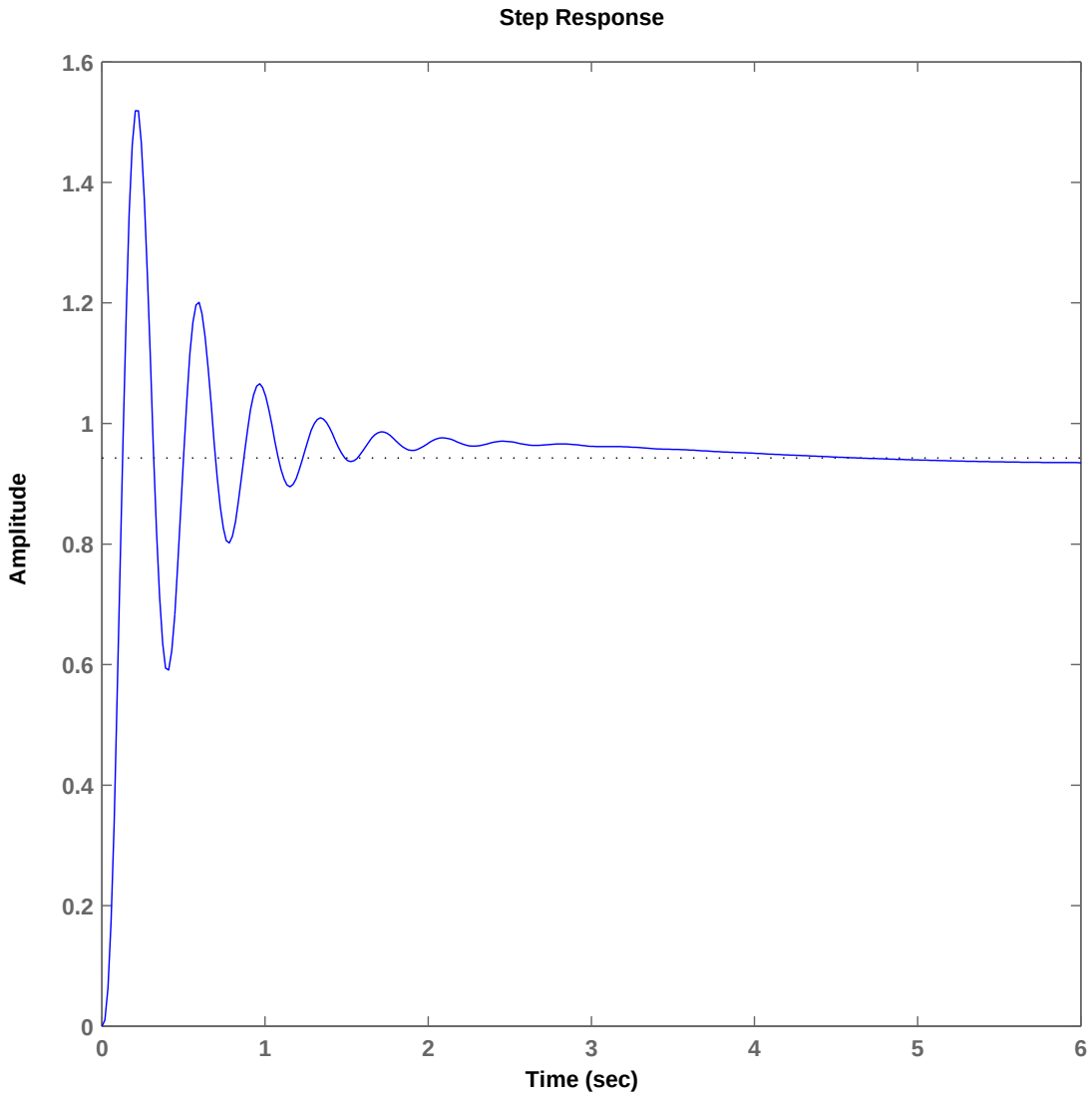
- Using `rlocfind`, we pick off the point

$$k = 10$$

- Remember to normalize!

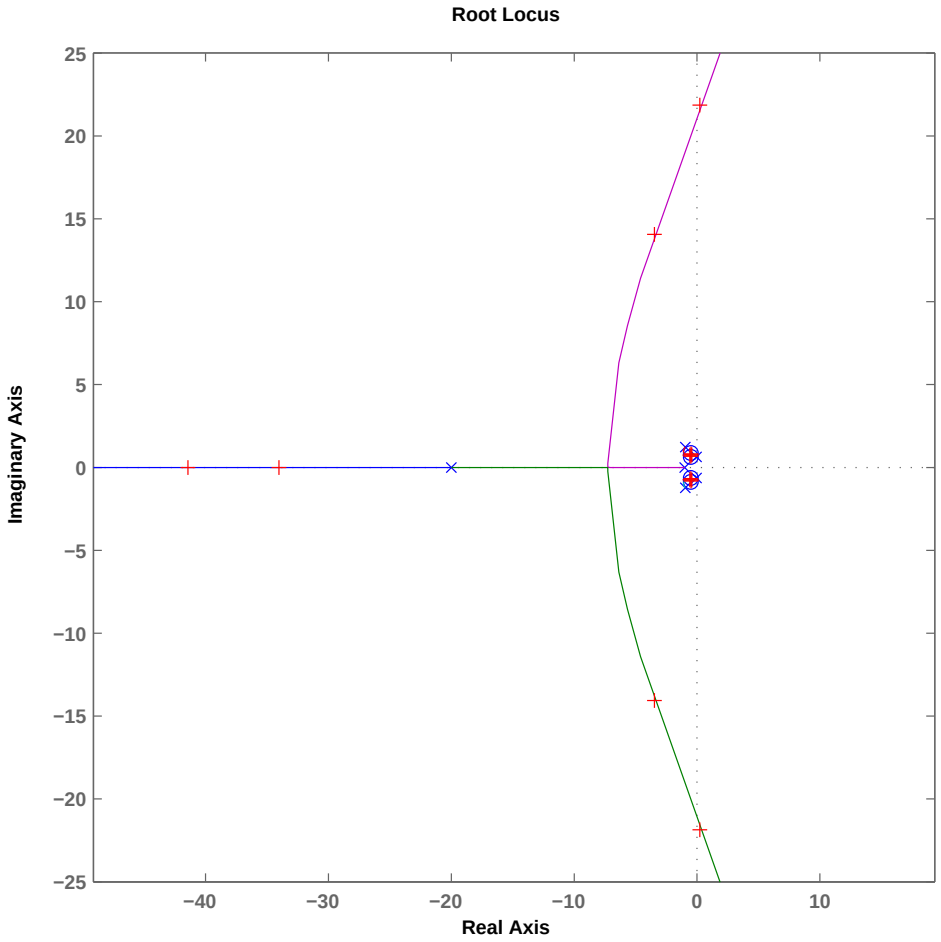


Notch-Filter Example



Notch-Filter Example

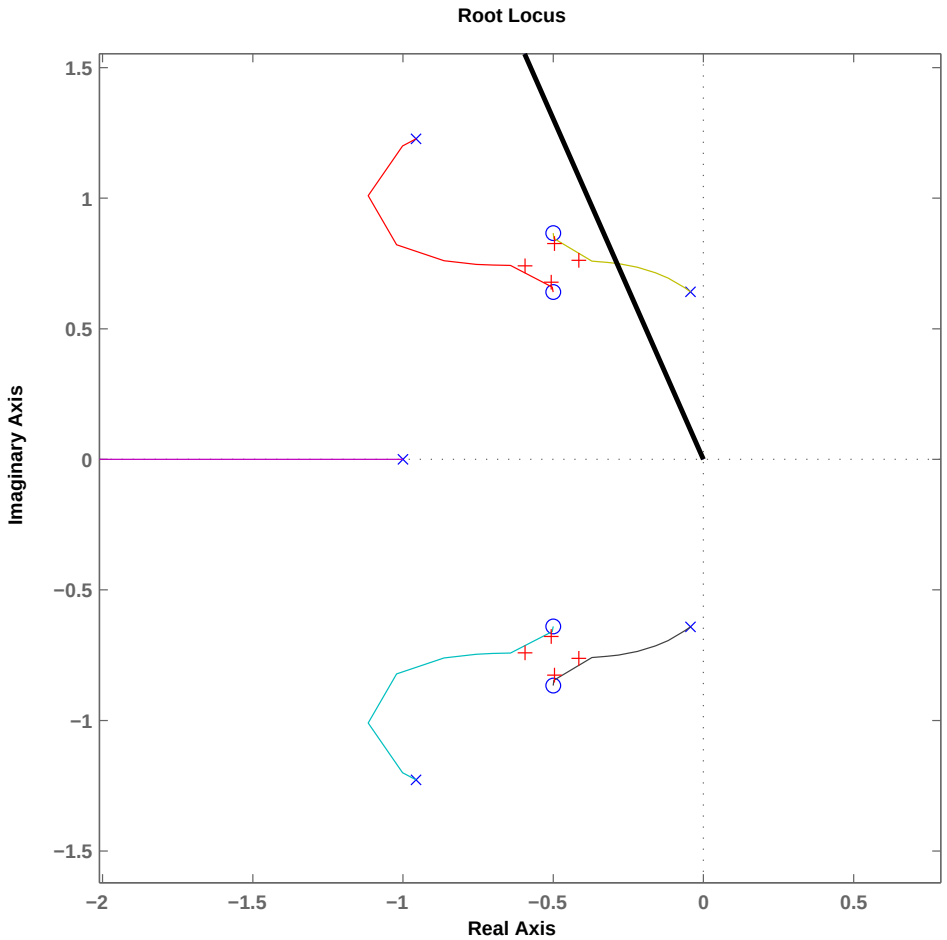
Don't Forget about the other poles!



$$k_{max} = 19.4$$

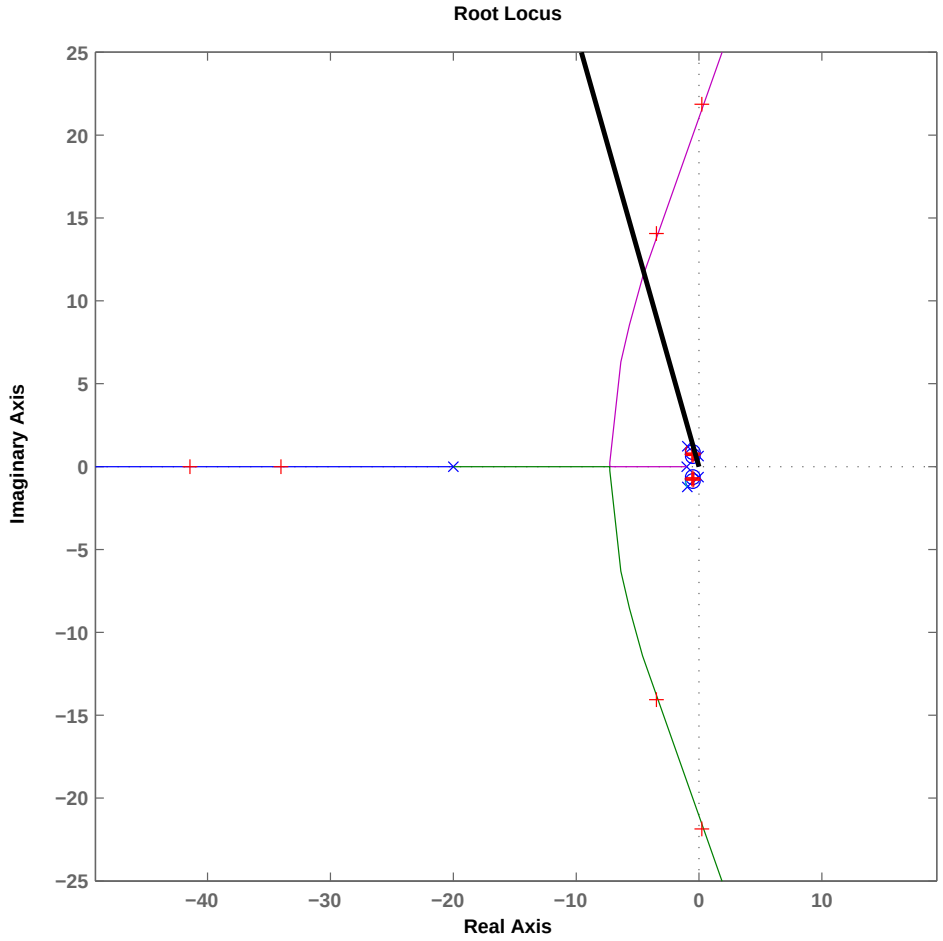
Notch-Filter Example

What about 30% overhoot?



Notch-Filter Example

What about 30% overhoot?



Don't forget those other poles!

Summary

What have we learned today?

Lead-Lag Compensation

- Designing Leads
  - Pole Placement using root locus
- Designing Lags
- Combining Leads and Lags

Notch Filters

- Providing complex zeros at targeted Locations
- Changes Departure Angles.

Next Lecture: The Frequency Domain